

Methods: PJ0209

ALMA Thesis Planner (automated pipeline)

0.1 0. Orientation and ground rules

Everything below uses only data from ADS/JAO.ALMA#2024.1.01396.S and standard ALMA/radio-astronomy tooling. You are not going to build a lens model. The scientific lever of this thesis is that, for emission that is co-spatial on the sky, magnification is a single multiplicative factor at a given image-plane position and therefore divides out of a line ratio formed *at that position*. That is what makes a resolved excitation study tractable in one year. The corollary — that differential magnification of components with different intrinsic sizes does *not* divide out — is a quantity you will measure and report, not a caveat you bury.

Two numbers you must not over-claim. First, the program's stated sample properties ($z = 1\text{--}3$, apparent $\log(L_{\text{IR}}) = 13\text{--}14 L_{\text{sun}}$, 3–4 transitions each drawn from $J = 2\text{--}15$ plus [CI], $0.1\text{--}0.2''$ synthesized beam, $>10\text{--}20$ beams across the emission) are design/sample-level statements from the proposal; treat them as expectations to be checked against the delivered data, not as measured properties of your source. Second, the "four orders of magnitude in relative high-J vs. mid-J line ratios" is a *sample-wide* spread quoted in the proposal. It is not a measurement of your target, and no sentence you write should imply that it is.

0.2 1. Data retrieval, inventory, and target selection (weeks 1–3)

Retrieve the delivered products for all eight program sources from the ALMA Science Archive: raw ASDMs, the pipeline calibration scripts ('scriptForPI.py' / pipeline 'casa_pipescript.py'), the calibrated measurement sets or the restoration path to regenerate them, the QA2 reports, and the delivered 'weblog'. Restore calibration with the CASA version recorded in the weblog — do not restore with a different major version, because the pipeline's flagging and 'statwt' behavior are version-dependent and you need bit-level reproducibility across the whole thesis.

Build an inventory table, one row per source per spectral window, recording: observed sky frequency coverage and channel width of each spw, integration time, number of executions, array configuration and baseline range (record the 5th and 80th percentile baseline lengths and the maximum recoverable scale), achieved continuum and line sensitivity from QA2, and the delivered synthesized beam major/minor axis and position angle per spw.

Line identification is a derived quantity, not a given. The data description does not state the redshift or the observed sky frequency of any individual transition for any source. For each source, take the redshift recorded in the delivered project metadata / observing setup, compute the expected observed frequency of each candidate transition from its rest frequency, verify against Splatalogue, and then confirm empirically that a line is actually detected at that sky frequency in the delivered spectra. Only after that empirical confirmation do you write down "this spw contains CO(J→J-1)". Record both the rest frequency and the observed sky frequency for every line in a

machine-readable table (‘lines.ecsv’) that the rest of the pipeline reads; never hard-code a frequency in a script.

Select the primary target on three delivered-data criteria, scored explicitly in a written selection memo: (i) peak and integrated S/N in the highest-J and lowest-J transitions (the ratio is only as good as its weakest line); (ii) completeness of J coverage actually delivered versus requested; (iii) morphology — you want extended arc structure or, ideally, multiple images, because that is what powers the internal consistency test in §5. Nominate a second source as backup and freeze the choice in writing at the end of week 3. All subsequent work is on the primary source.

0.3 2. Matched imaging — the technical heart (weeks 3–10)

The transitions span a wide range in J and therefore in sky frequency. Left alone, each cube would have a different native beam, a different uv coverage in units of wavelength, and a different largest recoverable angular scale. A ratio map built from mismatched cubes will show a spurious radial gradient — extended, low-excitation emission resolved out preferentially in the high-frequency (high-J) data would masquerade exactly as the compact-hot-core signal you are looking for. Removing this systematic *by construction* is the single most important technical act of the thesis.

2.1 Common uv range. Compute the uv distance distribution of each transition’s data in units of $k\lambda$, not meters. Define a common annulus: inner radius = the largest of the per-line minimum uv distances; outer radius = the smallest of the per-line maximum uv distances (in practice, use robust percentiles rather than absolute extrema so a handful of outlier baselines does not throw away sensitivity). Apply this identically to every transition via the ‘uvrange’ parameter in ‘tclean’, expressed in $k\lambda$. Document the fraction of visibilities discarded per line.

2.2 Common imaging parameters. Image every transition with the same cell size, same image size, same phase center, same Briggs robust value, and the same deconvolution setup (‘specmode=’cube’, ‘outframe=’LSRK’, multiscale scales chosen from the beam size and the source extent, ‘pbcor’ performed as a separate step so you retain both flat-noise and pb-corrected products). Choose the cell to give ~ 5 pixels across the minor axis of the *target* common beam.

2.3 Common circular beam. Determine the target restoring beam as a circle with FWHM slightly larger (≈ 5 –10%) than the largest native major axis among the transitions after the common- uv cut. Reach it two ways and compare: (a) ‘uvtaper’ in ‘tclean’ plus ‘restoringbeam’ set to the common circular beam; (b) native imaging followed by ‘imsmooth’ with ‘kernel=’gauss’, ‘targetres=True’. Path (b) is the safer default because the convolution is exact and traceable; path (a) is a cross-check. The two must agree at the few-percent level on integrated flux in a common aperture — if they do not, stop and find out why before proceeding.

2.4 Residual scaling (JvM). Cubes at different frequencies have different dirty-beam-to-clean-beam volume ratios, so un-scaled residuals contribute a different effective flux unit to each map. Compute the JvM epsilon factor for each cube and apply the residual rescaling consistently, or clean deeply enough (to $\approx 1\sigma$ with a well-defined mask) that the residual contribution is negligible. Whichever you choose, apply it identically to all transitions and record the epsilon values.

2.5 Continuum handling. Identify line-free channels from the delivered spectra per spw, fitting a low-order (0th or 1st) polynomial. Subtract in the uv plane with ‘uvcontsub’ (this is the primary path) and, as a cross-check, in the image plane. Retain and image the continuum itself as a deliverable product — you will need it both as a morphological reference and because continuum-subtraction choice is one of the systematics in §6. Produce a second, alternative continuum-

subtracted set using a deliberately different line-free channel selection and a different polynomial order; keep it for the systematics test rather than discarding it.

2.6 Common velocity and spatial grid. Regrid every cube onto an identical LSRK velocity axis with a common channel width (choose something like $30\text{--}50\text{ km s}^{-1}$, coarse enough for per-channel S/N, fine enough to resolve the line profile), using ‘imreframe’ and ‘imregrid’. Then regrid all cubes onto a single common spatial grid with ‘imregrid’ so that pixel (i,j) means the same sky position in every transition. Verify by cross-correlating the continuum-subtracted moment-0 maps: any systematic centroid offset between transitions larger than $\sim 1/10$ beam must be understood (check per-execution astrometry and phase-calibrator solutions) before you form ratios.

2.7 Primary beam. The primary beam FWHM scales inversely with frequency, so the high-J maps have the smallest field of view and the steepest response gradient. Apply primary-beam correction to all cubes and then mask the analysis region to where the primary-beam response exceeds a fixed threshold (0.5 is a reasonable floor; 0.8 is safer) *in every transition simultaneously*. Use the same mask for all lines.

Deliverable of this stage: a set of continuum-subtracted, primary-beam-corrected, matched-*uv*, matched-circular-beam, common-grid cubes plus a matched continuum image, all generated by a single scripted pipeline with parameters in one YAML/JSON configuration file.

0.4 3. Moment maps and noise (weeks 10–13)

Measure the noise empirically per channel per cube: take the RMS in signal-free regions of the flat-noise (pre-pb-correction) image, then propagate the primary-beam correction to obtain a spatially varying σ map. Confirm the noise is Gaussian and that channel-to-channel correlation is what you expect from the gridding kernel and any Hanning smoothing.

Define the velocity integration window from the highest-S/N transition, using a spatially integrated spectrum over a generous aperture, and apply that *identical* window to every transition to form moment-0 maps (‘immoments’, ‘moments=[0]’, no clipping inside the window — clipped moments bias low-S/N pixels). As a cross-check only, also build per-line windows tuned to each transition’s own profile; the two versions bound any bias from velocity-dependent excitation.

Propagate the uncertainty analytically: $\sigma_{\text{M0}} = \sigma_{\text{chan}} \times \Delta v \times \sqrt{N_{\text{chan}}}$, with σ_{chan} taken from the same channels used in the window, plus a term for correlated channels if Hanning smoothing was applied. Produce a σ_{M0} map alongside every moment-0 map. Also produce moment-1 and moment-2 maps for each transition; the velocity-gradient field is needed later, because the LVG column density is per unit velocity gradient and you should know whether the local line width varies across the source.

0.5 4. Ratio maps (weeks 13–16)

Form pixel-wise ratio maps for every available high-J/mid-J pair, and CO/[CI] where [CI] is delivered, working in units of integrated line flux density (Jy km s^{-1}) and, separately, in brightness-temperature-equivalent units (K km s^{-1}) after applying the ν^{-2} conversion — state clearly in every figure caption which convention a given map uses, since the two differ by a large frequency-dependent factor and mixing them is the classic error in this subfield.

Mask conservatively: a ratio pixel is retained only where *both* constituent maps exceed a fixed threshold (start at 5σ ; show the map at 3σ and 7σ as well to demonstrate that the structure is

not a masking artifact). Because the pixels are oversampled relative to the beam, also produce a beam-sampled version (one independent sample per beam area) for any statistical statement about gradients.

Uncertainties on the ratio combine the propagated σ_{M0} terms with the ALMA absolute flux-calibration uncertainty, which is frequency-dependent and larger in the higher bands. Treat the flux-calibration term as a *systematic* that scales the whole ratio map coherently, and quote it separately from the pixel-wise statistical error — it moves the ratio map up or down as a block, it does not create or destroy a gradient.

Present the results as: (i) a ratio-map atlas with S/N contours overlaid; (ii) ratio versus continuum-surface-brightness and ratio versus projected distance from the continuum peak, with beam-sampled points and error bars; (iii) a global ratio measured in a single large aperture, for direct comparison with the already-constrained global gas properties for this source.

0.6 5. Internal consistency test using the source’s own structure (weeks 14–16)

If the target is multiply imaged, identify image pairs/counterparts by their morphology and, where available, by matched velocity structure in the moment-1 maps — this is a morphological correspondence, not a lens model, and you should say so. Regions that correspond to the same intrinsic material must yield the same line ratio within the combined statistical and systematic errors, because magnification cancels in each independently. Any discrepancy is a direct, empirical measurement of your reduction’s residual bias, and it sets the floor on how small a ratio gradient you are entitled to call real. If the source is a single extended arc rather than multiply imaged, run the same test between positions along the arc that are symmetric about the arc’s fold, and be explicit that the test is weaker in that configuration.

0.7 6. Systematics work package (runs throughout; written up weeks 20–26)

Four tests, each a scripted re-run of the full pipeline with one thing changed, each reported as a difference map against the fiducial ratio map:

1. **Imaging weighting.** Re-run at two additional Briggs robust values and, if S/N allows, with natural weighting. The scatter across weightings at fixed common beam bounds the deconvolution-driven error.
2. **Residual uv mismatch.** Tighten the common uv annulus by a further 10–20% at both ends and re-derive the ratio maps. If the gradient survives, spatial filtering is not producing it.
3. **Continuum subtraction.** Compare the fiducial uv -plane subtraction against the image-plane subtraction and against the alternative line-free-channel/polynomial-order choice from §2.5.
4. **Differential magnification.** Using the multiply-imaged or arc structure from §5, quantify how much apparent ratio variation is induced purely by the fact that a more compact component is magnified differently from an extended one. Frame this concretely: given the observed positional spread of the counterimages and the measured size difference between the compact and extended components in your own maps, what fractional ratio change would that alone produce? This is a bounded, data-driven statement, and it must be carried into the interpretation of every ratio gradient you claim.

0.8 7. Region-based non-LTE excitation modelling (weeks 16–26)

Do not fit every pixel. Define ~5–10 apertures on physical grounds — the continuum/line peak, two or three positions along the arc, one or two outskirt regions, and one global aperture — each at

least one beam across, each drawn on the fiducial maps and then frozen before any fitting. Record aperture positions and sizes in a region file that is version-controlled.

For each aperture, extract integrated line fluxes for all transitions over the common velocity window, with uncertainties that include the aperture-correlated noise (scale the per-pixel σ by $\sqrt{N_{\text{pixels}}/N_{\text{pixels_per_beam}}}$, not by $\sqrt{N_{\text{pixels}}}$) and the flux-calibration systematic.

Model each aperture with a one-zone escape-probability / large-velocity-gradient non-LTE calculation (RADEX, or a Python wrapper such as pyradex/myRadex, using the code’s bundled CO and C collisional rate coefficients). Grid, coarsely and logarithmically:

- kinetic temperature T_{kin} , roughly 10–500 K,
- H_2 volume density $n(\text{H}_2)$, roughly 10^2 – 10^7 cm^{-3} ,
- CO column per unit velocity gradient N_{CO}/dv , roughly 10^{15} – $10^{19} \text{ cm}^{-2} (\text{km s}^{-1})^{-1}$.

Fit **line ratios**, not absolute luminosities. This is deliberate: the beam-filling factor and the magnification both cancel in a ratio taken within a single aperture, so you avoid introducing either as a free nuisance parameter. Include the CMB at the source redshift as the background radiation field in the escape-probability calculation — at $z = 1$ – 3 this is not negligible for the low-J lines, and omitting it biases the inferred excitation. Where [CI] is available, include the $\text{CO}/[\text{CI}]$ ratio, and state the assumed carbon-to-CO abundance ratio explicitly as an assumption with the value written in the text; show how the inferred parameters move if you vary it by a factor of a few.

With only 3–4 transitions per source you have 2–3 independent ratios and three parameters. Therefore the deliverable is a *likelihood surface*, not a best-fit point. Compute χ^2 (or a Gaussian log-likelihood) on the full grid, then either report the grid directly or refine with an MCMC (emcee) using flat priors over the grid ranges. For each aperture produce: the full corner plot / marginalized posteriors, and the 2D T_{kin} – $n(\text{H}_2)$ surface with the thermal-pressure product $n \cdot T$ overplotted as a family of curves. In most apertures $n \cdot T$ (or the pressure-like combination) will be far better constrained than either parameter alone — say so, and quantify it. Explicitly identify which apertures’ data do and do not break the T_{kin} – $n(\text{H}_2)$ degeneracy, and why (which J-values are bracketing the ladder turnover in that aperture). This is a legitimate, reportable result.

Finally, run the identical machinery on the global aperture and compare the resulting parameter region against the previously constrained global gas properties for this source. The scientific question is answered by the comparison of the per-aperture surfaces against each other and against the global one: if the apertures’ allowed regions are mutually inconsistent at the level set by your §5 and §6 systematics floors, the global ratio is a luminosity-weighted average over real internal gradients; if they overlap, the source is consistent with a spatially uniform excitation state at the precision the data support.

0.9 8. Software practice and reproducibility (throughout)

Everything is a script. Structure the repository as: ‘config/’ (one YAML per source with spectral setup, line table, uv range, target beam, velocity grid), ‘pipeline/’ (CASA imaging stage, moment stage, ratio stage), ‘models/’ (RADEX grid generation and fitting), ‘figures/’ (plotting only, reading from stored FITS and ECSV files). Use ‘astropy’, ‘spectral-cube’, ‘radio-beam’, ‘reproject’, ‘numpy’, ‘matplotlib’, and ‘emcee’; store all intermediate products as FITS with complete headers and all tables as ECSV. Pin the CASA and Python environment versions in the repository. The pipeline must be able to run end-to-end from calibrated measurement sets to final likelihood

surfaces with a single command and a config file, because the stated deliverable is a pipeline the parent program can point at the remaining program sources.

Include the required acknowledgements verbatim in every product and in the thesis: the ALMA acknowledgement naming ADS/JAO.ALMA#2024.1.01396.S, and the standard NRAO acknowledgement.

0.10 9. Contingency: pipeline validation on simulated cubes

If the delivered data for the selected source are not accessible at the thesis start, do **not** substitute a different dataset. Instead, build synthetic cubes from the program’s own stated observing parameters — a 0.1–0.2'' circular beam and a source spanning >10–20 beams — using CASA ‘simobserve’/‘simanalyze’ with antenna configurations that reproduce those design values at each of the relevant sky frequencies. Inject a known two-component model: a compact, highly excited component embedded in an extended, lower-excitation reservoir, with input T_{kin} , $n(\text{H}_2)$, and N_{CO}/dv chosen from your RADEX grid, and arranged into an arc-plus-counterimage morphology so that the §5 consistency test and the §6 differential-magnification test can both be exercised. Run the identical §2–§7 pipeline on these cubes and demonstrate quantitatively that (a) the input ratio gradient is recovered when matched imaging is applied, and (b) a spurious gradient appears when the common- uv or common-beam step is deliberately skipped — that second demonstration is itself a useful validation figure. Then switch to the real data the day it is delivered, with the pipeline already frozen and tested.